

Juno Kim

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Profile

I am a Ph.D. student in Computer Science at the University of California, Berkeley, advised by Profs. Jason D. Lee and Song Mei. My interests lie in the **mathematical foundations of deep learning**, with a focus on nonconvex optimization, dynamical analysis, and computational-statistical guarantees for neural networks. I am also interested in understanding and harnessing the mathematical reasoning capabilities of foundation models.

I am interning at Google DeepMind during summer 2026, hosted by Hossein Mobahi.

Previously, I received my B.Sc. in Mathematics and Statistics at Seoul National University as Valedictorian of '23, and my M.Sc. in Mathematical Informatics at the University of Tokyo advised by Prof. Taiji Suzuki, where I received the Dean's Award for outstanding research.

Publications

■ selected works, *equal contribution

Nicholas Barnfield, **Juno Kim**, Eshaan Nichani, Jason D. Lee, Yue M. Lu. Sharp Capacity Thresholds in Linear Associative Memory: From Winner-Take-All to Listwise Retrieval. *Under review*.

■ **Juno Kim***, Eshaan Nichani*, Denny Wu, Alberto Bietti, Jason D. Lee. Sharp Capacity Scaling of Spectral Optimizers in Learning Associative Memory. *ICML 2026 HiLD Workshop (Oral)*.

Juno Kim, Jihun Yun, Jason D. Lee, Kwang-Sung Jun. Coverage Improvement and Fast Convergence of On-policy Preference Learning. **ICML 2026**.

Stanley Wei, **Juno Kim**. Provable Benefits of RLVR over SFT for Reasoning Models: Learning to Backtrack Efficiently. **ICML 2026**.

Jihun Yun*, **Juno Kim***, Jongho Park, Junhyuck Kim, Jongha Jon Ryu, Jaewoong Cho, Kwang-Sung Jun. Alignment as Distribution Learning: Your Preference Model is Explicitly a Language Model. *Under review*.

Anming Gu*, **Juno Kim***. Mirror Mean-Field Langevin Dynamics. **ICML 2026**.

Juno Kim, Denny Wu, Jason D. Lee, Taiji Suzuki. Metastable Dynamics of Chain-of-Thought Reasoning: Provable Benefits of Search, RL and Distillation. **ICML 2025**.

Naoya Yamamoto, **Juno Kim**, Taiji Suzuki. Hessian-guided Perturbed Wasserstein Gradient Flows for Escaping Saddle Points. **NeurIPS 2025**.

■ **Juno Kim**, Taiji Suzuki. Transformers Provably Solve Parity Efficiently with Chain of Thought. **ICLR 2025 (Oral)**.

Juno Kim, Dimitri Meunier, Arthur Gretton, Taiji Suzuki, Zhu Li. Optimality and Adaptivity of Deep Neural Features for Instrumental Variable Regression. **ICLR 2025**.

Juno Kim, Tai Nakamaki, Taiji Suzuki. Transformers are Minimax Optimal Nonparametric In-Context Learners. **NeurIPS 2024** and *ICML 2024 TF2M Workshop (Best Paper Award)*.

■ **Juno Kim**, Taiji Suzuki. Transformers Learn Nonlinear Features In Context: Nonconvex Mean-field Dynamics on the Attention Landscape. **ICML 2024 (Oral)**.

Juno Kim, Kakei Yamamoto, Kazusato Oko, Zhuoran Yang, Taiji Suzuki. Symmetric Mean-field Langevin Dynamics for Distributional Minimax Problems. **ICLR 2024 (Spotlight)**.

Juno Kim*, Jaehyuk Kwon*, Mincheol Cho*, Hyunjong Lee, Joong-Ho Won. t^3 -Variational Autoencoder: Learning Heavy-tailed Data with Student's t and Power Divergence. **ICLR 2024**.

Juno Kim, Otto van Koert. Hessian Based Smoothing Splines for Manifold Learning. arXiv preprint *arXiv:2302.05025*, 2023.

Juno Kim*, Yonghwan Kim*, Otto van Koert. Reeb Flows without Simple Global Surfaces of Section. *Involve: A Journal of Mathematics*, 15(5), pp. 813–842, 2022.

Education

University of California, Berkeley Aug 2025 – current
Ph.D. student in Electrical Engineering and Computer Sciences

University of Tokyo Apr 2023 – Mar 2025
M.Sc. in Mathematical Informatics (GPA 4.0/4.0)
Dean's Award for Research Achievement
Thesis: Statistical and Dynamical Analysis of Transformers: In-Context Learning and CoT Reasoning

Seoul National University Mar 2018 – Feb 2023
B.Sc. in Statistics & Mathematical Sciences (GPA 4.28/4.3)
Valedictorian of the College of Natural Sciences
Thesis: Token and Corpus Imputation in Statistical Language Modeling via Semantic Embeddings

Experience

Google DeepMind, Student Researcher 2026

KRAFTON AI, Research Intern 2025

RIKEN Center for Advanced Intelligence Project, Part-time Researcher Dec 2023 – Mar 2025

Simons Institute, UC Berkeley, Visiting Researcher Nov 2024 – Dec 2024

Gatsby Computational Neuroscience Unit, UCL, Visiting Researcher Aug 2024

Seoul National University, Undergraduate Research Intern Jun 2019 – Feb 2023

Seoul National University, Department of Statistics Peer Tutor Mar 2022 – Feb 2023

Invited Talks

- KAIST AI (Host: Chulhee Yun) Jul 14, 2026
- Tsinghua University (Host: Kaifeng Lyu) (online) May 15, 2026
- Flatiron Institute (Host: Alberto Bietti) May 15, 2026
- University of Pennsylvania (Host: Yuxin Chen) May 14, 2026
- Kempner Institute, Harvard University (Host: Bingbin Liu) May 13, 2026
- Carnegie Mellon University (Host: Andrej Risteski) May 11, 2026
- NLP Colloquium, Japan (Host: Sho Yokoi) (online) May 21, 2025
- Flatiron Institute (Host: Denny Wu) Mar 15, 2025
- Vector Institute (Host: Anastasis Kratsios) (online) Jan 24, 2025

Reviewer

- AISTATS'24, ICML'24,25,26 (Gold Reviewer), NeurIPS'24,25,26, ICLR'25 (Notable Reviewer),26

Honors & Awards

- Dean's Award for Research Achievement, IST, University of Tokyo 2025
- Doctoral Course (DC1) Research Fellowship, JSPS Declined
- Japanese Government (MEXT) Scholarship Apr 2023 – Mar 2025
- President Award, Highest Honors, Seoul National University 2023
- President Award, Korean Statistical Society 2023
- National Scholarship, Kwanjeong Educational Foundation Mar 2020 – Feb 2023
- 4th Place, Simon Marais Mathematics Competition 2022
- Eminence Scholarship, Seoul National University Sep 2018 – Feb 2020
- Gold Prize, College Mathematics Competition, Korean Mathematical Society 2019

Skills

Languages Korean: Native, English: Fluent (TOEFL 117), Japanese: Fluent (JLPT N1), German: Basic

Coding (Advanced) Python, PyTorch, R (Basic) C++, Java, MATLAB